F70LA Life Insurance Mathematics A (2025/26)
Tutorial 4 Solutions

1. With $v := 1.04^{-1}$,

   (a)
   $$ {}_5p_{55} = \frac{\ell_{60}}{\ell_{55}} = \frac{98\,484.31}{99\,176.23} = 0.99302. $$

   (b)
   $$ A^{1}_{55:\overline{5}|} = A_{55} - v^5\,{}_5p_{55}\,A_{60} = 0.29962 - 1.04^{-5}(0.99302)(0.35954) = 0.00617. $$

   (c)
   $$ A_{55:\overline{5}|} = A^{1}_{55:\overline{5}|} + v^5\,{}_5p_{55} = 0.00617 + 1.04^{-5}(0.99302) = 0.82236. $$

   (d) The EPV is
   $$ 50\,000\,A_{60} = 50\,000(0.35954) = £17\,997.00. $$

2. Using backward recursion, i.e. $A_x = v\,q_x + v\,p_x A_{x+1}$,

   $$ A_{60} = v\,q_{60} + v\,p_{60}A_{61} = 1.04^{-1}(0.002058) + 1.04^{-1}\,(1 - 0.002058)\,(0.37265) = 0.35956 $$

   and using the value of A_{60} calculated above,

   $$ A_{59} = v\,q_{59} + v\,p_{59}A_{60} = 1.04^{-1}(0.001801) + 1.04^{-1}\,(1 - 0.001801)\,(0.35956) = 0.34684. $$

3. We are given

   $$ A_{45} = 0.24850, \qquad A^{1}_{45:\overline{20}|} = 0.03205 \qquad A_{45:\overline{20}|} = 0.46276. $$

   and want to calculate A_{65}. First

   $$ A^{1}_{45:\overline{20}|} = A_{45} - v^{20}\,{}_{20}p_{45}\,A_{65} \quad \Rightarrow \quad A_{65} = \frac{A_{45} - A^{1}_{45:\overline{20}|}}{v^{20}\,{}_{20}p_{45}}. $$

   Therefore, we need to determine $v^{20}\,{}_{20}p_{45}$. Since

   $$ A_{45:\overline{20}|} = A^{1}_{45:\overline{20}|} + v^{20}\,{}_{20}p_{45} \quad \Rightarrow \quad v^{20}\,{}_{20}p_{45} = A_{45:\overline{20}|} - A^{1}_{45:\overline{20}|} = 0.46276 - 0.03205 = 0.43071, $$

   it follows that
   $$ A_{65} = \frac{0.24850 - 0.03205}{0.43071} = 0.50254. $$

4. The select life table is:

x	$\ell_{[x]}$	$\ell_{[x]+1}$	$\ell_{[x]+2}$	$\ell_{[x]+3}$	ℓ_{x+4}	$x+4$
40	100 000	99 899	99 724	99 520	99 288	44
41	99 802	99 689	99 502	99 283	99 033	45
42	99 597	99 471	99 268	99 030	98 752	46
43	99 365	99 225	99 007	98 747	98 435	47
44	99 120	98 964	98 726	98 429	98 067	48

With $v := 1.05^{-1}$,

(a)

$$\begin{aligned} A^{1}_{[40]:\overline{2}|} = \sum_{k=0}^{1} v^{k+1}{}_{k}|q_{[40]} =& v\, q_{[40]} + v^2 p_{[40]}\, q_{[40]+1} \\ =& v\left(1 - \frac{\ell_{[40]+1}}{\ell_{[40]}}\right) + v^2 \frac{\ell_{[40]+1}}{\ell_{[40]}}\left(1 - \frac{\ell_{[40]+2}}{\ell_{[40]+1}}\right) \\ =& v\left(1 - \frac{99\,899}{100\,000}\right) + v^2 \frac{99\,899}{100\,000}\left(1 - \frac{99\,724}{99\,899}\right) \\ =& 0.002549. \end{aligned}$$

(b)

$$A_{[41]+2:\overline{3}|} = A^{1}_{[41]+2:\overline{3}|} + v^3{}_3p_{[41]+2} = \sum_{k=0}^{2} v^{k+1}{}_{k}|q_{[41]+2} + v^3{}_3p_{[41]+2}.$$

As

$${}_{k}|q_{[41]+2} = {}_{k}p_{[41]+2}\, q_{[41]+2+k} = \frac{\ell_{[41]+2+k}}{\ell_{[41]+2}}\left(1 - \frac{\ell_{[41]+k+3}}{\ell_{[41]+2+k}}\right) = \frac{\ell_{[41]+2+k} - \ell_{[41]+k+3}}{\ell_{[41]+2}} = \frac{d_{[41]+2+k}}{\ell_{[41]+2}},$$

then

$$\begin{aligned} A_{[41]+2:\overline{3}|} =& \frac{v d_{[41]+2} + v^2 d_{[41]+3} + v^3 d_{45} + v^3 \ell_{46}}{\ell_{[41]+2}} \\ =& \frac{(99\,502 - 99\,283)v + (99\,283 - 99\,033)v^2 + (99\,033 - 98\,752)v^3 + (98\,752)v^3}{99\,502} \\ =& \frac{219v + 250v^2 + 281v^3 + 98\,752v^3}{99\,502} \\ =& 0.864141. \end{aligned}$$

5. (a) (i) Z_1 is the present value of a benefit payable immediately on the death of (x). The amount of the death benefit is 200 units if death occurs before age $x+20$ and 100 units if death occurs after age $x+20$.

(ii) Z_2 is the present value of a benefit of 100 units payable immediately either on the death of (x) if should death occurs between ages $x+8$ and $x+20$ or payable at age $x+20$ should (x) survive to that age.

(b) (i) Either

$$\mathbb{E}\left(Z_1\right) = 200\int_0^{20} v^t{}_t p_x \mu_{x+t} dt + 100\int_{20}^{\infty} v^t{}_t p_x \mu_{x+t} dt$$

or

$$\mathbb{E}\left(Z_1\right) = 100\int_0^{\infty} v^t{}_t p_x \mu_{x+t} dt + 100\int_0^{20} v^t{}_t p_x \mu_{x+t} dt.$$

(ii)

$$\mathbb{E}\left(Z_2\right) = 100\int_8^{20} v^t{}_t p_x \mu_{x+t} dt + 100 v^{20}{}_{20} p_x.$$

(c) (i) Either one of the expressions below.

$$\mathbb{E}\left(Z_1\right) = 200\bar{A}^1_{x:\overline{20|}} + 100\,{}_{20|}\bar{A}_x = 100\bar{A}_x + 100\bar{A}^1_{x:\overline{20|}}.$$

(ii) Either one of the expressions below.

$$\mathbb{E}\left(Z_2\right) = 100\,{}_{8|}\bar{A}^1_{x:\overline{12|}} + 100A_{x:\overline{20|}}^{\;\;\frac{1}{}} = 100\left(\bar{A}^1_{x:\overline{20|}} - \bar{A}^1_{x:\overline{8|}}\right) + 100A_{x:\overline{20|}}^{\;\;\frac{1}{}}.$$

In all expressions, you can replace the deferred notation and the pure endowment with actuarial notation of the form ${}_t p_x v^t$ (e.g. instead of ${}_{20|}\bar{A}_x$, you can write ${}_{20}p_x v^{20}\,\bar{A}_{x+20}$. Or instead of $A_{x:\overline{20|}}^{\;\;\frac{1}{}}$, you can write ${}_{20}p_x v^{20}$.

(d) Recall that the covariance of two random variables Z_1 and Z_2 is $\mathbb{C}ov(Z_1, Z_2) = \mathbb{E}\left(Z_1 Z_2\right) - \mathbb{E}\left(Z_1\right)\mathbb{E}\left(Z_2\right)$. We've already calculated $\mathbb{E}\left(Z_1\right)$ and $\mathbb{E}\left(Z_2\right)$, so it remains to calculate $\mathbb{E}\left(Z_1 Z_2\right)$. First,

$$Z_1 Z_2 = \begin{cases} (0)(200v^{T_x}) & \text{if } T_x \le 8, \\ (100v^{T_x})(200v^{T_x}) & \text{if } 8 < T_x \le 20, \\ (100v^{20})(100v^{T_x}) & \text{if } T_x > 20. \end{cases} = \begin{cases} 0 & \text{if } T_x \le 8, \\ 20\,000v^{2T_x} & \text{if } 8 < T_x \le 20, \\ 10\,000v^{20+T_x} & \text{if } T_x > 20. \end{cases}$$

Then

$$\mathbb{E}\left(Z_1 Z_2\right) = 20\,000\left({}^2\bar{A}^1_{x:\overline{20|}} - {}^2\bar{A}^1_{x:\overline{8|}}\right) + 10\,000v^{20}\left(\bar{A}_x - \bar{A}^1_{x:\overline{20|}}\right),$$

and, with ${}^2\bar{A}^1_{x:\overline{20|}}$ evaluated at the effective rate of interest i per annum equal in value to $\bar{A}^1_{x:\overline{20|}}$ evaluated at the effective rate of interest $(1+i)^2 - 1$ per annum,

$$\begin{aligned} \mathbb{C}ov\left(Z_1, Z_2\right) =& \mathbb{E}\left(Z_1 Z_2\right) - \mathbb{E}\left(Z_1\right)\mathbb{E}\left(Z_2\right) \\ =& 20\,000\left({}^2\bar{A}^1_{x:\overline{20|}} - {}^2\bar{A}^1_{x:\overline{8|}}\right) + 10\,000v^{20}\left(\bar{A}_x - \bar{A}^1_{x:\overline{20|}}\right) \\ & \left(100\bar{A}_x + 100\bar{A}^1_{x:\overline{20|}}\right)\left(100\,{}_{8|}\bar{A}^1_{x:\overline{12|}} + 100A_{x:\overline{20|}}^{\;\;\frac{1}{}}\right). \end{aligned}$$

6. (a) Let the random variable T_x represent the future lifetime of (x). First,

$$Z_1 = v^{T_x}\,\mathbb{1}_{[T_x \le n]} \qquad \text{and} \qquad Z_2 = v^{T_x}.$$

The covariance is

$$\mathbb{C}ov(Z_1, Z_2) = \mathbb{E}\left((Z_1 - \mathbb{E}(Z_1))\ (Z_2 - \mathbb{E}(Z_2))\right) = \mathbb{E}\left(Z_1\, Z_2\right) - \mathbb{E}(Z_1)\,\mathbb{E}(Z_2).$$

We know

$$\mathbb{E}(Z_1) = \bar{A}^{1}_{x:\overline{n}|} \qquad \text{and} \qquad \mathbb{E}(Z_2) = \bar{A}_x.$$

As

$$Z_1\, Z_2 = v^{2T_x}\,\mathbb{1}_{[T_x \le n]} \qquad \Rightarrow \qquad \mathbb{E}\left(Z_1\, Z_2\right) = {}^{2}\bar{A}^{1}_{x:\overline{n}|}$$

then

$$\mathbb{C}ov(Z_1, Z_2) = {}^{2}\bar{A}^{1}_{x:\overline{n}|} - \bar{A}^{1}_{x:\overline{n}|}\,\bar{A}_x.$$

(b) Let the random variable T_y represent the future lifetime of (y). Then

$$Z_3 = v^{T_y}.$$

The covariance $\mathbb{C}ov(Z_1, Z_3) = 0$. This is because:

- Z_1 is a function of the random variable T_x,
- Z_3 is a function of the random variable T_y, and
- T_x and T_y are independent random variables.
- Therefore, as functions of independent random variables inherit the independence, the covariance of Z_1 and Z_3 is zero.